

Qatar-2b

R-2176

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_120214 · created 2026-08-13T01:02:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455971.91806 +/- 0.00049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.043 +/- 0.067
Transit depth (Rp/Rs)^2	0.18 +/- 0.57 [%]
Orbital Inclination (inc)	89.56 +/- 1.74
Airmass coefficient 1 (a1)	2.3 +/- 0.44
Airmass coefficient 2 (a2)	-0.67 +/- 0.45
Scatter in the residuals of the lightcurve fit is	19.77 %
Best Comparison Star	#3 - [105, 456]
Optimal Method	PSF photometry
Transit Duration (day)	0.0571 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.043 ± 0.067 vs 0.16327 (deviation 1.8σ)
Duration fit vs published	0.0571 d vs 0.0767 d (deviation 9.78σ)
Mid-time O-C	0.9 min
Depth SNR	0.6
Residual scatter	19.77 %

Light curve

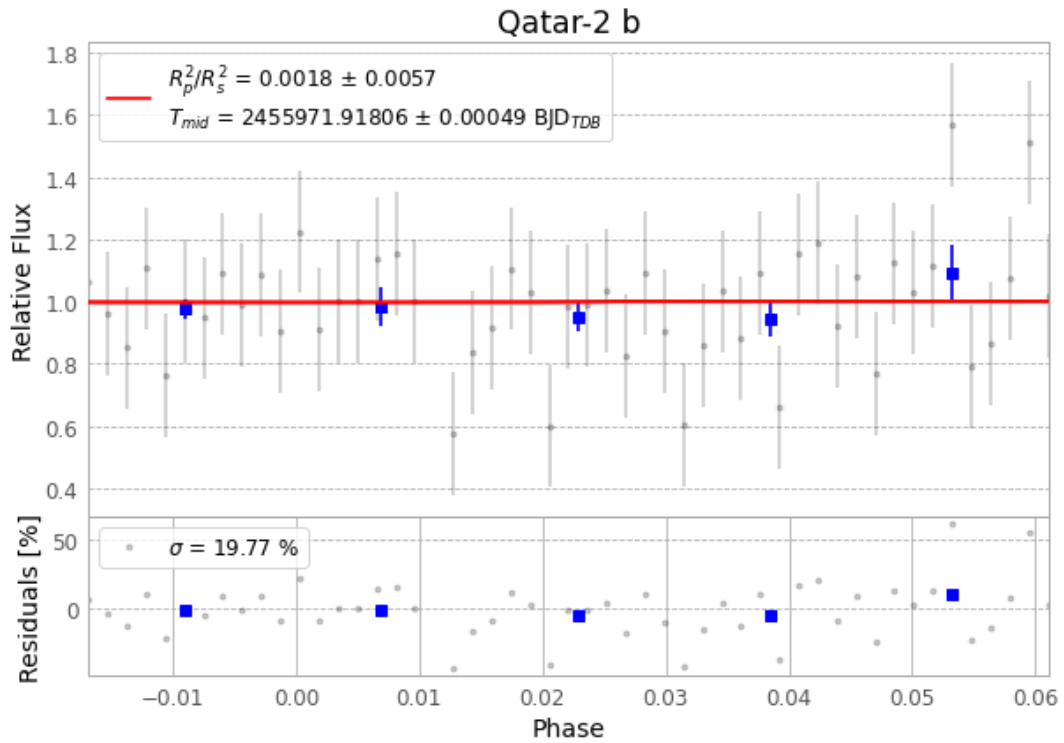


Figure 1 — FinalLightCurve_Qatar-2 b_2012-02-14.png (SHA-256 1373634ef251fd0f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [239, 345], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 497931830e7eeabb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

50 FITS frames (33 MB), Qatar-2120214092414.FITS → Qatar-2120214115413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-120214.log (ab78e85b76e6...), FinalLightCurve_Qatar-2 b_2012-02-14.png (1373634ef251...), FinalLightCurve_Qatar-2 b_2012-02-14.csv (40c6f9a3287b...)

Compute resources

Wall time 148.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 01:02Z.